

Retail Customer Insights & Segmentation Analysis

Python • MySQL • Power BI

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

Attribute	Details
Source	Kaggle – Public Retail Transactions Dataset
Rows	3,900
Columns	18
Missing Data	37 values in Review Rating column

Key Features:

- Customer demographics: Age, Gender, Location, Subscription Status
- Purchase details: Item Purchased, Category, Purchase Amount, Season, Size, Color
- Shopping behavior: Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

3. Exploratory Data Analysis using Python

Data preparation and cleaning was performed in Python using Pandas:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.
- **Missing Data Handling:** Imputed missing values in the Review Rating column using the category-specific median — not an overall average — to preserve data accuracy.
- **Column Standardization:** Renamed all columns to snake_case for consistency and database compatibility.
- **Feature Engineering:** Created `age_group` column by binning customer ages into Young Adult, Adult, Middle-aged, and Senior. Created `purchase_frequency_days` column from frequency text values.
- **Data Consistency Check:** Verified that `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used` to avoid duplication.

- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using MySQL

Structured analysis was performed in MySQL to answer 14 business questions across revenue, segmentation, discount behaviour, product performance, and seasonal trends.

Q1. Revenue by Gender

```
SELECT gender, SUM(purchase_amount) AS revenue FROM customer GROUP BY gender;
```

Gender	Revenue
Female	\$75,191
Male	\$157,890

Q2. High-Spending Discount Users

```
SELECT customer_id, purchase_amount FROM customer WHERE discount_applied = 'Yes' AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

Result
839 customers identified

Q3. Top 5 Products by Average Review Rating

```
SELECT item_purchased, ROUND(AVG(review_rating), 2) AS average_product_rating FROM customer GROUP BY item_purchased ORDER BY AVG(review_rating) DESC LIMIT 5;
```

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Q4. Shipping Type vs Average Purchase Amount

```
SELECT shipping_type, ROUND(AVG(purchase_amount), 2) AS avg_purchase_amount FROM customer WHERE shipping_type IN ('Standard', 'Express') GROUP BY shipping_type;
```

Shipping Type	Avg Amount
Standard	\$58.46
Express	\$60.48

Q5. Subscribers vs Non-Subscribers Spend

```
SELECT subscription_status, COUNT(customer_id) AS total_customers, ROUND(AVG(purchase_amount), 2) AS avg_spend, ROUND(SUM(purchase_amount), 2) AS total_revenue FROM customer GROUP BY subscription_status ORDER BY total_revenue DESC, avg_spend DESC;
```

Status	Customers	Avg Spend	Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

Q6. Discount-Dependent Products

```
SELECT item_purchased, ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS discount_rate FROM customer GROUP BY item_purchased ORDER BY discount_rate DESC LIMIT 5;
```

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Q7. Customer Segmentation (CTE)

```
WITH customer_type AS ( SELECT customer_id, previous_purchases, CASE WHEN previous_purchases = 1 THEN 'New' WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning' ELSE 'Loyal' END AS customer_segment FROM customer ) SELECT customer_segment, COUNT(*) AS number_of_customers FROM customer_type GROUP BY customer_segment;
```

Segment	Count
Loyal	3,116
Returning	701
New	83

Q8. Top 3 Products per Category (Window Function)

```
WITH item_counts AS ( SELECT category, item_purchased, COUNT(customer_id) AS total_orders, ROW_NUMBER() OVER ( PARTITION BY category ORDER BY COUNT(customer_id) DESC ) AS item_rank FROM customer GROUP BY category, item_purchased ) SELECT item_rank, category, item_purchased, total_orders FROM item_counts WHERE item_rank <= 3;
```

Rank	Category	Product	Orders
1	Accessories	Jewelry	171
1	Clothing	Blouse	171
1	Footwear	Sandals	160
1	Outerwear	Jacket	163

Q9. Repeat Buyers vs Subscription Status

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers FROM customer WHERE previous_purchases > 5 GROUP BY subscription_status;
```

Status	Repeat Buyers
No	2,518
Yes	958

Q10. Revenue by Age Group

```
SELECT age_group, SUM(purchase_amount) AS total_revenue FROM customer GROUP BY age_group ORDER BY total_revenue DESC;
```

Age Group	Revenue
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

Q11. Discount Utilization Rate by Age Group

```
SELECT age_group, ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS discount_rate FROM customer GROUP BY age_group ORDER BY discount_rate DESC;
```

Age Group	Discount Rate (%)
Senior	44.17
Adult	43.10
Young Adult	43.09
Middle-aged	41.68

Q12. Revenue & Transactions by Season

```
SELECT season, COUNT(customer_id) AS total_transactions, ROUND(AVG(purchase_amount), 2) AS avg_purchase_amount, ROUND(SUM(purchase_amount), 2) AS total_revenue FROM customer GROUP BY season ORDER BY total_revenue DESC;
```

Season	Total Transactions	Avg Purchase Amount	Total Revenue
Fall	975	\$61.56	\$60,018
Spring	999	\$58.74	\$58,679
Winter	971	\$60.36	\$58,607
Summer	955	\$58.41	\$55,777

Q13. Segment vs Discount Usage (CTE)

```
WITH segmented AS ( SELECT customer_id, purchase_amount, discount_applied, CASE WHEN
previous_purchases = 1 THEN 'New' WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
ELSE 'Loyal' END AS customer_segment FROM customer ) SELECT customer_segment,
COUNT(customer_id) AS total_customers, ROUND(AVG(purchase_amount), 2) AS avg_purchase_amount,
ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
discount_usage_rate FROM segmented GROUP BY customer_segment ORDER BY avg_purchase_amount DESC;
```

Segment	Total Customers	Avg Purchase Amount	Discount Usage Rate (%)
Returning	701	\$60.93	40.80
Loyal	3,116	\$59.54	43.65
New	83	\$58.47	37.35

Q14. Category Rating Rank (Window Function)

```
SELECT category, ROUND(AVG(review_rating), 2) AS avg_rating, ROUND(SUM(purchase_amount), 2) AS
total_revenue, RANK() OVER (ORDER BY AVG(review_rating) DESC) AS rating_rank FROM customer
GROUP BY category;
```

Category	Avg Rating	Total Revenue	Rating Rank
Footwear	3.79	\$36,093	1
Accessories	3.77	\$74,200	2
Outerwear	3.75	\$18,524	3
Clothing	3.72	\$104,264	4

5. Dashboard in Power BI

An interactive Power BI dashboard was built to present findings visually to business stakeholders.

- KPI Cards: 3.9K Customers | \$59.76 Average Purchase Amount | 3.75 Average Review Rating
- Donut Chart: Subscription Status breakdown — 27% Subscribed, 73% Not Subscribed
- Bar Charts: Revenue by Category and Sales by Category
- Horizontal Bars: Revenue by Age Group and Sales by Age Group
- Slicers: Gender, Category, Shipping Type, Subscription Status — enabling full dynamic filtering
- DAX Measures: Custom calculations for customer count, avg spend, and avg rating

6. Key Findings

- **Loyalty Gap:** 2,518 repeat buyers (more than 5 purchases) are NOT subscribed vs only 958 who are — the biggest business opportunity identified in the project.
- **Gender Revenue Split:** Male customers generate more than double the revenue of female customers (\$157,890 vs \$75,191), suggesting gender-targeted campaigns.
- **Discount Dependency:** Hat (50%), Sneakers (49.66%), and Coat (49.07%) have the highest discount rates — nearly half of all purchases use a discount, raising margin concerns.

- **Subscription Spend Parity:** Subscribers and non-subscribers spend almost the same on average (\$59.49 vs \$59.87), meaning subscriptions are not driving higher spend — a key insight for re-evaluating the subscription value proposition.
- **Top Performing Segment:** Young Adults contribute the highest revenue (\$62,143), making them the primary target demographic for campaigns.
- **Category Performance:** Clothing leads in both revenue and sales volume; Outerwear is the lowest performer across both metrics.

7. Business Recommendations

- **Convert Loyal Non-Subscribers:** Target the 2,518 repeat buyers who are not subscribed with exclusive membership benefits — this is the highest-impact growth lever identified.
- **Review Discount Policy:** Products like Hat and Sneakers have ~50% discount rates. Investigate whether discounts are driving incremental sales or simply eroding margin on purchases that would happen anyway.
- **Boost Subscription Value:** Since subscribers and non-subscribers spend the same on average, the subscription program needs stronger incentives — exclusive early access, free shipping, or loyalty points.
- **Targeted Marketing by Age:** Young Adults drive the most revenue. Focus digital marketing spend on this segment while designing re-engagement campaigns for Senior customers who contribute the least.
- **Product Positioning:** Promote top-rated products (Gloves, Sandals, Boots) prominently in campaigns. High ratings combined with strong sales create natural social proof.
- **Seasonal Campaigns:** Use seasonal revenue data to plan inventory and promotional budgets ahead of peak seasons rather than reacting after the fact.